01 / 03

Choose the position from the business exposure

A currency hedge starts with a business cash flow, not an opinion about exchange rates. A U.S. company expecting foreign-currency revenue is exposed to that currency weakening. A company with a foreign-currency bill is exposed to that currency strengthening. The derivative should offset that exposure.

Business cash flow	Adverse move	Forward position
Foreign-currency receipt	Foreign currency weakens	Sell foreign currency
Foreign-currency payment	Foreign currency strengthens	Buy foreign currency

Let S_T be spot at maturity and F the contracted forward rate, both in USD per unit of foreign currency. Let N be the foreign-currency amount. A long forward buys that currency; a short forward sells it. These payoffs are in USD.

$$\text{Long payoff}_T = N(S_T - F)$$

$$\text{Short payoff}_T = N(F - S_T)$$

If the foreign currency becomes more expensive, the long forward gains because it lets the holder buy at the lower contracted rate. The short forward loses for the same reason. The two sides have equal and opposite payoffs.

Evaluate the business and hedge together

For a receivable, the spot-converted receipt is N x S_T. Adding a short-forward payoff leaves N x F. For a payable, subtracting the long-forward gain from the spot purchase cost again leaves N x F. The uncertain spot-rate terms cancel.

A derivative loss is not necessarily a failed hedge
The derivative can lose precisely when the underlying exposure improves. The relevant outcome is the combined cash flow, not the derivative profit on its own.

02 / 03

A receivable with three possible exchange rates

A U.S. manufacturer will receive AUD 1 million in three months and sells it forward at AUDUSD = 0.64. This is an assumed contract quote to illustrate the payoff; we are not deriving 0.64 from a specified interest-rate curve.

AUDUSD at receipt	Receipt at spot	Short-forward payoff	Combined USD
0.68	680,000	-40,000	640,000
0.65	650,000	-10,000	640,000
0.60	600,000	+40,000	640,000

At 0.68, the stronger Australian dollar increases the operating receipt. The forward loses USD 40,000 because the company agreed to sell the currency below spot. At 0.60, the smaller operating receipt is offset by a positive forward payoff. In both cases, the hedge preserves USD 640,000.

$$NS_T + N(F - S_T) = NF$$

Forwards and futures create different cash timing

A forward is typically an over-the-counter agreement with a specified amount and settlement date. A futures contract uses standardized terms and is marked to market through the clearing system. Daily gains and losses generate variation-margin cash flows before the business payment arrives.

A company can have an economically sensible futures hedge and still need cash for margin calls. Contract-size rounding, mismatched expiration dates and differences between the contract reference and the actual exposure can leave residual risk.

What the constant-total example assumes
Amount, currency and date match exactly. The example ignores transaction costs, default and the financing effect of intermediate margin cash flows so that we can see the hedge mechanism clearly.

Reference: CME Clearing, "Money Calculations for Futures and Options," on settlement variation and daily marking to market.
cmegroup.com/clearing/files/CME-Money-Calculations-Futures-and-Options.pdf

Price each hedge, then evaluate the portfolio

For a quote in USD per unit of foreign currency, covered interest parity places the USD growth factor in the numerator and the foreign growth factor in the denominator. The following example uses annual simple rates.

$$F = S\frac{1 + r_{\text{USD}}T}{1 + r_{\text{foreign}}T}$$

A firm will receive EUR 500,000 and pay GBP 300,000 in six months. Spot EURUSD is 1.0850 and GBPUSD is 1.2700. Annual simple rates are USD 5.25%, EUR 3.75%, and GBP 5.00%. The firm sells its future euros and buys its future pounds.

Exposure	Six-month forward	Locked USD amount
Receive EUR 500,000	EURUSD 1.092988	546,493.87 receipt
Pay GBP 300,000	GBPUSD 1.271549	381,464.63 payment
Net USD cash flow	Receipt less payment	165,029.23

Calculate each hedge in its own currency before taking the net dollar amount. A euro receipt does not directly cancel a pound payment merely because both occur on the same date. Their dollar values respond to different exchange rates.

The hedge makes the budget more predictable

Once exchange rates are fixed, management can evaluate the operating decision without relying on a favorable currency move. A plain forward gives up the benefit of a favorable move in exchange for protection against an unfavorable one.

Zero initial value does not mean zero risk

A newly agreed fair forward normally has approximately zero initial value. It can acquire positive or negative value as prices change. Credit exposure, collateral obligations and cancellation of the underlying business transaction remain relevant even when no option premium was paid.

A useful sequence
Identify the business exposure, choose the offsetting position, calculate a consistent forward quote, and compare combined cash flows across scenarios. Tie the payoff sign to the position you actually hold.